

Methodology Notes - Master Evidence Packet (ASCII-Safe Format)

These notes explain how each chart and comparison in this project was created. All text has been converted into ASCII-safe formatting so it renders cleanly in all PDF readers, including government systems.

MN1 - Data Processing Rules

Data arrives on different schedules and in different forms. Some agencies publish numbers every month, others only once a year. To compare these series on the same chart, the dates must be cleaned and aligned. The goal is to match the period being measured, not the release date.

First, all dates are converted into a simple year-month format. Daily or weekly values are averaged into a single number for that month. Quarterly values are assigned to the first month of the quarter. Annual values are treated as the value for the end of the year.

After alignment, any rows with empty values or duplicate timestamps are removed. No attempt is made to guess missing numbers. When charting, quarterly series may appear as steps or flat sections where the original data has fewer updates.

Different agencies sometimes use different cutoff dates for a reporting month, so their January values may not cover the exact same days. Aligning by the period measured is still the most transparent option. Anyone can verify this by downloading the same CSV files from official sources and checking the date fields.

MN2 - Inflation Adjustment Methods

A dollar from the past does not have the same buying power as a dollar today. To compare wages, rents, or debt fairly over time, values are adjusted for inflation. This project distinguishes between nominal values (the amount at the time) and real values (the amount in today's dollars).

The basic method is direct. A base year is chosen. The Consumer Price Index for All Urban Consumers (CPI-U) is used. For a past value, the CPI for the base year is divided by the CPI for the earlier year. The result is multiplied by the past amount to convert it into current dollars.

For worker-specific comparisons, the CPI for wage earners (CPI-W) may be used. This is relevant for Social Security cost-of-living adjustments and certain wage discussions.

Inflation adjustment has limits. The CPI represents an average basket of goods. If someone spends more of their budget on rent or healthcare than the CPI assumes, their personal inflation may be higher. Even with this limitation, CPI is the standard public measure and allows for reproducible real-dollar comparisons.

MN3 - Indexing and Normalization Rules

Many charts compare data series with different units, such as home prices in dollars and wages in dollars per hour. To make them comparable, both are converted to index values. Indexing sets a starting point to a value of 100, and all later values are expressed relative to that starting point.

The formula is simple: current value divided by the starting value, multiplied by 100. If a series doubles, its index moves from 100 to 200. If it rises by fifty percent, the index moves from 100 to 150.

Indexing highlights growth rates and relative change. It makes it easy to see whether home prices, rents, or wages grew faster over the same period.

Indexing also has limitations. It hides the absolute starting level. Index charts are usually paired with charts showing dollar amounts when the absolute values matter.

MN4 - Housing Methodology

Housing stress is measured by rent burden and affordability. Rent burden asks how much of a household income is spent on rent. Federal standards say housing should not exceed 30 percent of gross income. Above 30 percent is considered burdened. Above 50 percent is severely burdened.

Rent burden is calculated by dividing annual rent by annual income and multiplying by 100. If the value exceeds 30, the typical household is burdened. To calculate the income needed to afford rent safely, annual rent is divided by 0.30.

Two main rent sources are used. Census tables show what current renters pay. Real-time listing indexes show what new renters would pay today. Home price indexes from federal agencies track price changes using repeat sales to avoid distortions from changing home sizes or features.

Limitations include data lag and incomplete coverage. Census data can be over a year behind. Listing-based rent indexes may exclude smaller landlords. These methods may understate recent rent spikes.

MN5 - Monetary and Financial Methodology

Monetary and financial charts focus on how borrowing costs and debt obligations affect households. Key measures include interest rates, debt service ratios, and central bank liquidity indicators such as the Federal Reserve balance sheet.

Debt service ratios measure how much of disposable income is required to make scheduled debt payments. There are separate ratios for total debt, mortgage debt, and consumer debt. These ratios answer the question of how much of the national paycheck is already committed to lenders.

Interest-rate charts use benchmark series for mortgages, credit cards, auto loans, and personal loans. Weekly or quarterly data is averaged into monthly values so it aligns with monthly inflation or wage data.

Liquidity charts track the Federal Reserve balance sheet, reverse repurchase agreements, and the Treasury General Account. Rising values generally indicate looser monetary conditions. Falling values show tightening. These charts do not identify direct beneficiaries, but they show when financial conditions are shifting.

MN6 - Congressional Analysis Methods

The congressional analysis compares what officials say with how they vote. Narratives come from hearings, speeches, and press releases. Behavior is documented through roll-call votes. The focus is on documented alignment or contradiction between statements and actions.

Soundbites are identified by scanning transcripts for recurring phrases such as now is not the time, blank check, or we have no choice. Each instance is tagged with a date, speaker, and topic. These tags build summary tables and heatmaps showing when certain arguments increase in frequency.

Flip tables match specific quotes with specific votes. For a flip entry, there must be a clear quote and a related bill. If a member publicly opposes a policy but later votes for it, that pairing is documented with timestamps and vote numbers.

This method relies exclusively on public records. It does not capture private motives, off-record remarks, or negotiations. It shows only the public record of statements and actions.

MN7 - Limitations and Interpretation

Charts show patterns, timing, and direction, but they cannot prove intent or causation. When two lines move together over time, this suggests a relationship, but it does not prove that one caused the other.

Government data has limits. Many series are revised after release. Housing and income statistics may lag behind current conditions. National averages smooth over local extremes.

Readers should pay attention to overall trends, not just single spikes. The axes on a chart show whether values represent dollars, percentages, or index levels. These details affect interpretation.

Every chart in this project can be reproduced from public sources. Series codes, table codes, and links are provided in the Data Sources packet so readers can download the same numbers and rebuild the visuals. Transparent method notes allow anyone to examine or question the analysis rather than relying on trust.